



Blackbox Speaker

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Abstract

Rigid Audio Tuning

Portable audio devices are essential to modern daily life. However, consumer Bluetooth speakers are typically restricted to static factory-preset tuning curves designed for a single generic environment.

This rigidity results in poor, unadaptable playback quality in challenging acoustics, such as small resonant spaces or wide outdoor environments.

The App-Dependency Issue

Existing premium alternatives (e.g., Sonos, Bose, Anker) restrict custom equalization solely to proprietary smartphone companion apps.

If app updates break, or when pairing with non-mobile hosts, custom sound profiles are lost entirely. End users lack physical, immediate tactile control of their audio spectrum.

Methods and Materials

A2DP Sink Stack

Manages wireless device pairing and receives live SBC-encoded audio streams on a high-priority FreeRTOS task, ensuring zero-latency buffering.

Real-time DSP

Implements second-order IIR biquad filters to process digital audio in real time. Features adjustable gain and dynamic cutoff frequencies for bass/treble shelf filters.

UI Interaction

Utilizes Hardware Pulse Counters (PCNT) for rotary encoders. Toggles between gain and frequency modes via interrupt-driven Finite State Machine (FSM).

I2S/I2C Audio Pipe

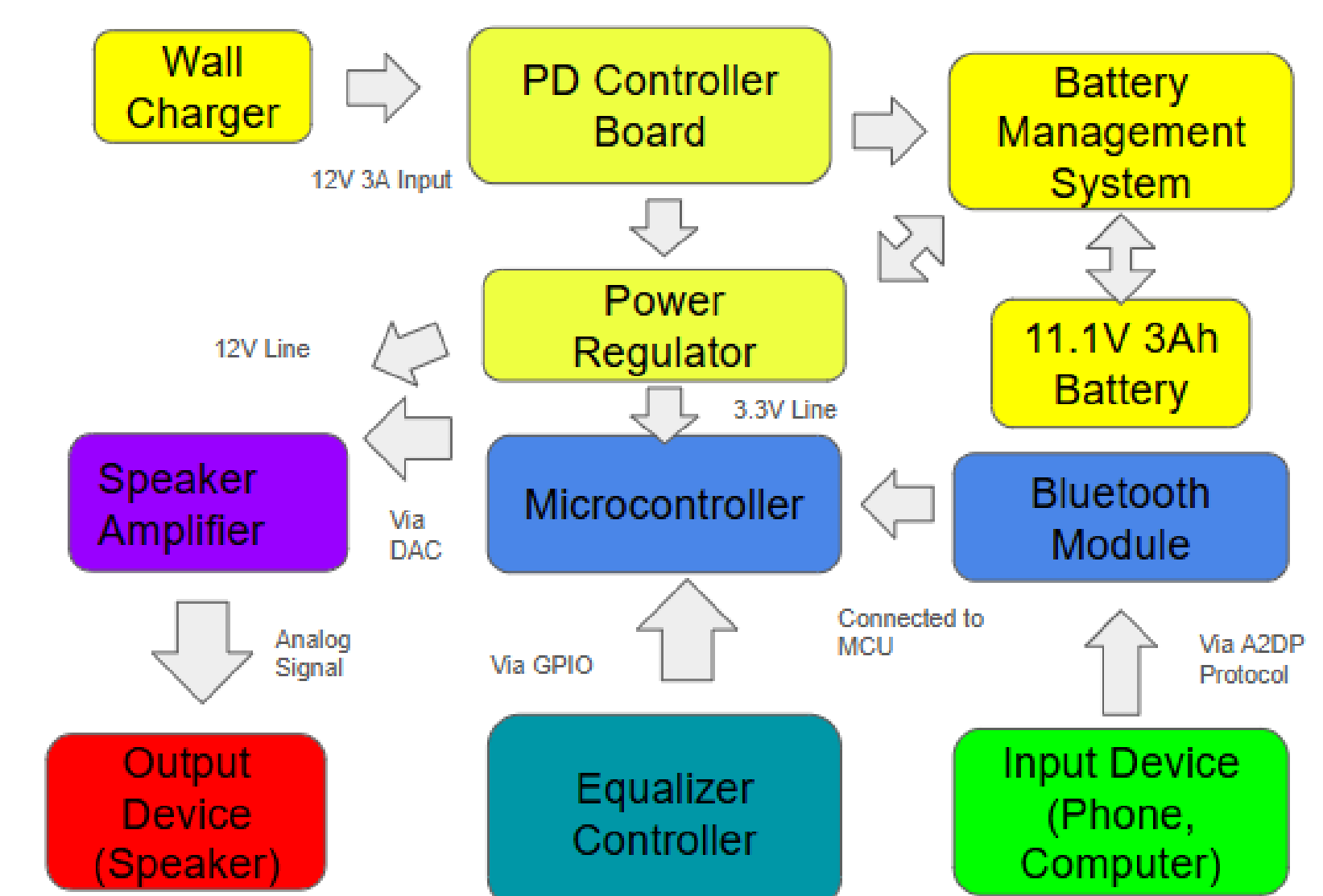
Leverages High-Precision Audio PLL (APLL) for I2S transmission to the TAD5112 DAC. I2C protocol manages register initialization and battery health monitoring.



ESP32: The system brain: Dual-core 240MHz MCU managing Bluetooth A2DP streams and high-speed real-time DSP calculations.

TAD5112: High-fidelity 32-bit DAC from TI. Converts filtered digital PCM into high-dynamic-range analog differential signals.

Overall System Design



Background

The Audio Customization Gap

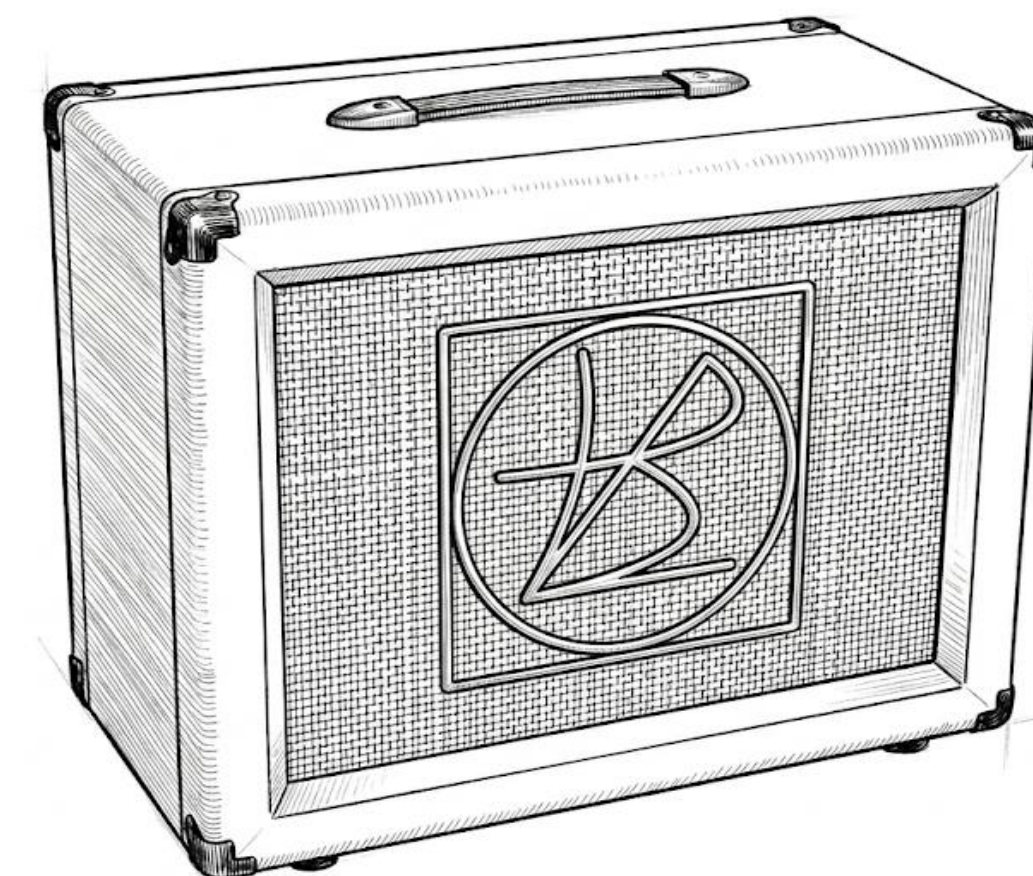
Commercial portable speakers are historically built with rigid, factory-tuned acoustic curves. This project bridges that gap by returning direct processing agency back to the hardware layer.

-  **Acoustic Rigidity:** Preset curves fail to adapt to real-world environments, resulting in muddy low-ends in tight spaces or thin audio outdoors.
-  **App-Dependency Flaw:** Existing custom EQs rely on software companion apps. If updates break or hosts change, custom parameters are completely lost.
-  **Standalone Solution:** Moving computation to the edge by running biquad DSP filters directly on the ESP32. This guarantees immediate, tactile control right from the chassis.

References

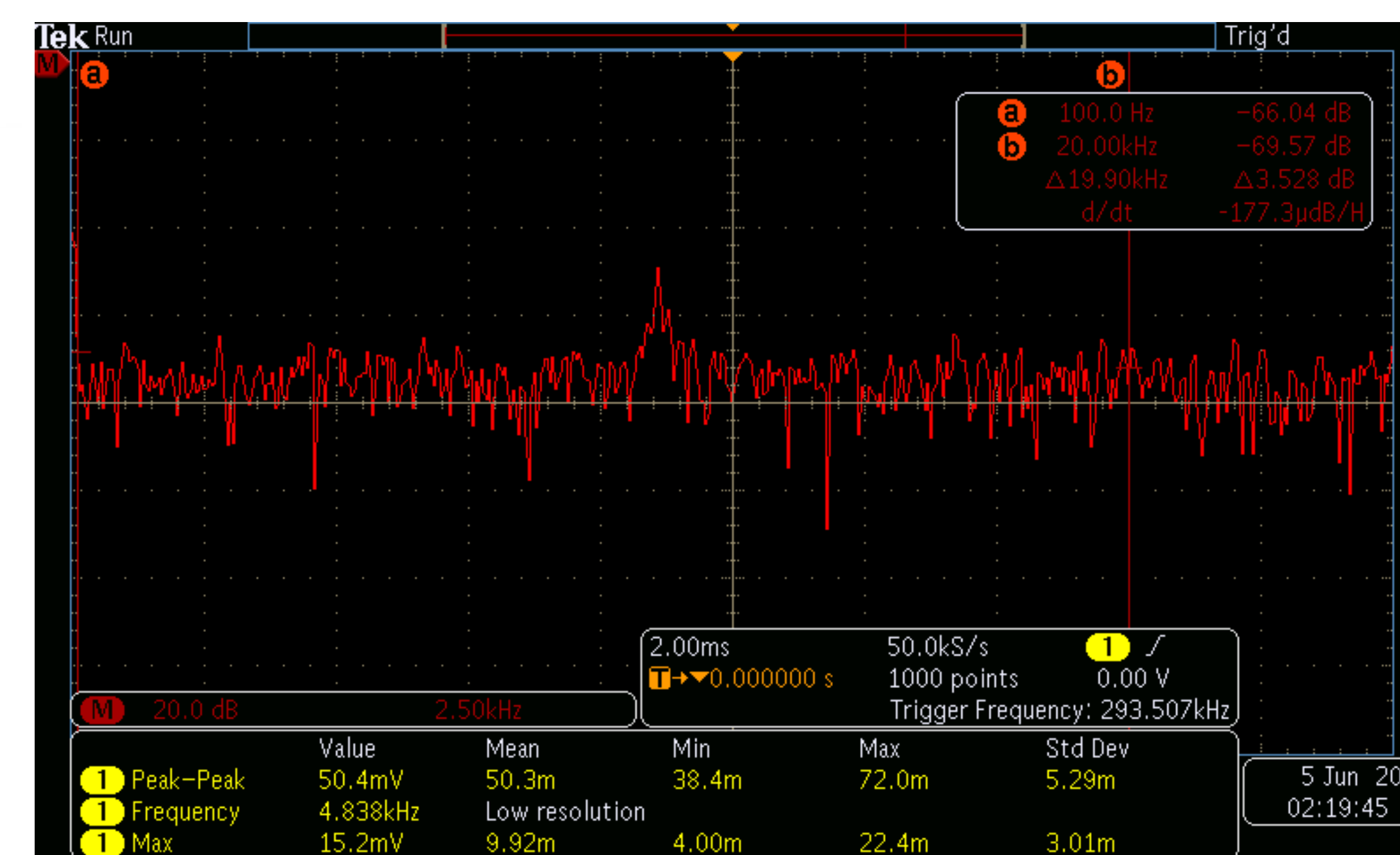
Microcontroller (ESP32): Espressif Systems. "ESP32 Modules and Boards." *ESP-IDF Programming Guide (v3.1.5)*.
Digital-to-Analog Converter (TAD5112): Texas Instruments. "TAD5112 Performance Audio DAC." *TI Semiconductor Product Catalog*.
Rotary Encoder (PEC11R): Bourns Inc. "PEC11R-4320K-S0012 Mechanical Incremental Encoder." Distributed by DigiKey.
External Amplifier Module (HiLetgo Board): Texas Instruments / HiLetgo. "TPA3116 100W Mono Channel Digital Audio Power Amplifier High-Power AMP Module." Distributed by Amazon. Product Documentation

Results



- Semi-battery powered Bluetooth speaker with custom equalizer.
- Low noise floor of -65 dB to -70 dB for audible frequency bands of 20Hz to 20KHz.
- Our designed amp configuration was incorrect on the PCB so we had to order a prebuilt mono channel amplifier board.
- Issues with the ground loops between the PCB and amplifier board that created audible noise. This was fixed with a separation of power stage.

```
I (1252486) A2DP_DSP: [ENC1] Mode: BASS | Bass: -0.2 dB @ 208 Hz | Treble: 0.0 dB @ 4860 Hz
I (1253706) A2DP_DSP: [ENC1] Mode: TREBLE | Bass: -0.2 dB @ 208 Hz | Treble: 0.0 dB @ 4860 Hz
I (1254806) A2DP_DSP: [ENC1] Mode: TREBLE | Bass: -0.2 dB @ 208 Hz | Treble: -0.2 dB @ 4860 Hz
I (1257146) A2DP_DSP: [ENC1] Mode: BASS | Bass: 0.0 dB @ 208 Hz | Treble: -0.2 dB @ 4860 Hz
I (1258486) A2DP_DSP: [ENC1] Mode: BASS | Bass: -0.2 dB @ 208 Hz | Treble: -0.2 dB @ 4860 Hz
I (1265206) A2DP_DSP: [ENC2] Mode: BASS_FREQ | Bass cutoff: 204 Hz | Treble cutoff: 4860 Hz
I (1267106) A2DP_DSP: [ENC2] Mode: TREBLE_FREQ | Bass cutoff: 204 Hz | Treble cutoff: 5000 Hz
I (1268446) A2DP_DSP: [ENC2] Mode: TREBLE_FREQ | Bass cutoff: 204 Hz | Treble cutoff: 4860 Hz
```



- Selection of a mid-frequency band in addition to bass and treble bands.
- Quarter inch port in for connecting instruments to the amplifier input.
- Implementation of more rotary dials for finer audio customizations.
- Addition of LCD display showing the following:
- Wave form after it passes through the filters
- Gain of frequency bands
- Range of frequency bands
- Bluetooth connection indication
- Current battery life

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Capstone Project
Electrical & Computer Engineering Program